

Factories Overview

singlecell_factory + multiomics_r_factory technical reference

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Table of contents

section
1. Executive summary
2. Architecture
3. Capability matrix
4. Bundle v2.1 contract
5. Workflow walkthroughs
6. scfactory CLI reference
7. NC2024 claim coverage
8. Test coverage and quality gates
9. Known limitations and roadmap
Appendix: file index

1. Executive summary

singlecell_factory is a Python/AnnData heavy-compute pipeline organized as 33 concrete module files across the layers ingest, quality_control, latent_structure, markers, annotation, state_dynamics, biology, aggregation, reproduction, spatial, multimodal, genomic_optional, external_validation, covariates, and reporting. The catalog declares 29 active module specs (3 mandatory + 26 optional) with explicit dependencies, modality tags, and bridge_ready flags. The CLI entrypoint is `scfactory run`.

multiomics_r_factory is the R-side downstream visualization workspace. It consumes the bundle v2.1 contract emitted by `scripts/export_singlecell_r_bundle.py`, reads it via `R_bundle/io_bundle.R`, and renders Seurat-based plots via the `R/*_module.R` files. Together: 97 collected tests on the named contract surface, 4 starter recipes, 3 modality extensions (protein/spatial/multimodal_obs), and an NC2024 paper-faithful reproduction lane.

Who should use it: pipeline users via `scfactory run`; agents via the documented bundle v2.1 contract; paper-faithful reproductions via `--recipe nc2024_paper`.

2. Architecture

Python (singlecell_factory)

R (multiomics_r_factory)

Python layer	modules	bridge	R module
ingest	2	1 of 2	annotation_module
quality_control	2	0 of 2	batch_integration_module
latent_structure	2	1 of 2	cli_utils
covariates	1	0 of 1	composition_plots
markers	2	2 of 2	dim_plots
annotation	1	1 of 1	expression_plots
state_dynamics	4	0 of 4	integration_module
genomic_optional	2	0 of 2	io_bridge
biology	6	3 of 6	marker_module
external_validation	1	0 of 1	pipeline_steps
reporting	1	1 of 1	preprocessing_module
aggregation	1	1 of 1	protein_module
reproduction	1	1 of 1	qc_plots
spatial	2	2 of 2	spatial_module
multimodal	1	1 of 1	theme_config

Bridge: bundle v2.1

schema_version: singlecell_r_bundle_v2.1 (back-compat: ...v2)

extensions: protein, spatial, multimodal_obs

claim_guard: not_for_de_or_new_quantitative_claims_...

atomicity: temp dir + os.replace + sha256 + size cross-check

Architecture: Python heavy-compute on the left writes the bundle v2.1 manifest in the middle; R reads it on the right. Bridge_ready Python modules are the ones whose outputs are intended for R-side consumption. The bundle contract is the only stable agent handoff surface between the two repos.

3. Capability matrix

Auto-extracted from workflow/modular/module_catalog.py (29 modules).

module	layer	modality	depends_on	bridge	status
cellranger	ingest	singlecell_rna	-	no	stable
qc	quality_control	singlecell_rna	cellranger	no	stable
doublet_detection	quality_control	singlecell_rna	qc	no	stable
clustering	latent_structure	singlecell_rna	doublet_detection	yes	stable
cell_cycle	covariates	singlecell_rna	clustering	no	stable
batch_correction	latent_structure	singlecell_rna	clustering	no	stable
differential_expression	markers	singlecell_rna	clustering	yes	stable
annotation	annotation	singlecell_rna	clustering	yes	stable
trajectory	state_dynamics	singlecell_rna	clustering	no	stable
pseudo_velocity	state_dynamics	singlecell_rna	trajectory	no	stable
rna_velocity	state_dynamics	singlecell_rna_splicing	clustering	no	stable
cnv_inference	genomic_optionals	copy_number	clustering	no	stable
pathway_analysis	biology	singlecell_rna	differential_expression	no	stable
cell_communication	biology	singlecell_rna	annotation	no	stable
gene_regulatory_network	biology	singlecell_rna	clustering	no	stable
validate_cbiportal	external_validation	external_cancer_genomics	differential_expression	no	stable
immune_phenotyping	biology	singlecell_rna	annotation	yes	stable
tumor_microenvironment	biology	singlecell_rna	annotation	yes	stable
gene_signature_scoring	biology	singlecell_rna	clustering	yes	stable
evolution	genomic_optionals	copy_number_state_dynamics	cnv_inference, trajectory	no	stable
pseudobulk_de	markers	singlecell_rna	differential_expression	yes	stable
cell_fate	state_dynamics	singlecell_rna	trajectory	no	stable
composition	reporting	singlecell_rna	annotation	yes	stable
metacell	aggregation	singlecell_rna	clustering	yes	stable
paper_repro	reproduction	singlecell_rna	clustering	yes	stable
protein_adt	ingest	protein_adt	qc	yes	stable
spatial_ingest	spatial	spatial_transcriptomics	qc	yes	stable
spatial_neighborhoods	spatial	spatial_transcriptomics	spatial_ingest	yes	stable
multimodal_integration	multimodal	multimodal_joint_embedding	clustering	yes	experimental

4. Bundle v2.1 contract

field / topic	value
schema_version (v2.1)	singlecell_r_bundle_v2.1
schema_version (back-compatible)	singlecell_r_bundle_v2
accepted by ExportConfig	v1, v2, v2.1
top-level manifest	schema_version, source, bundle, cell_alignment, expression, files, extensions
KNOWN_EXTENSION_KEYS	protein, spatial, multimodal_obsm
protein extension	CLR-normalized ADT (parquet); fields: normalization, isotype_controls
spatial extension	(x,y) coords + library metadata (parquet); image data is STRING-pointer only
multimodal_obsm extension	EXPERIMENTAL; one parquet per joint embedding (X_wnn, X_mofa)
claim_guard literal	not_for_de_or_new_quantitative_claims_without_full_object_validation

CLAIM_GUARD explanation. The literal string is surfaced in R as both a `message()` call and as `attr(expr_sparse, "claim_guard")`. It means bundle data is plotting-only and must not be used for new quantitative claims (DE, thresholding, etc.) without going back to the full `AnnData` and re-validating.

Atomicity. Writers stage every file under a `TemporaryDirectory`, compute SHA256 per file (recorded in the manifest, cross-checked against on-disk byte size), and write the manifest last. `os.replace(temp_dir, final_output_dir)` performs the atomic swap; any pre-existing target is moved aside as a backup and restored on failure.

5. Workflow walkthroughs

5.nc2024_paper

NC2024 NSCLC paper-faithful baseline reproduction

Command:

```
python scripts/scfactory.py run --recipe nc2024_paper <input>
```

optional module
clustering
differential_expression
annotation
trajectory
paper_repro

env var	value
SC_CLUSTERING_ENGINE	sparse_exact

bundle field	value
enabled	False
include_protein	False
include_spatial	False
include_multimodal_obsm	False
schema_version	v2.1

When to use: Honors NC2024 paper-aligned parameters. Uses sparse-exact clustering (Phase 7+ default for ~300k+ cohorts). Bundle is OFF by default — `paper_repro` emits its own parity artifacts; pass `--bundle` explicitly if you also want the v2.1 R bundle.

5.quick_explore

Fast first-look exploration: clustering + DE only, bundle on

Command:

```
python scripts/scfactory.py run --recipe quick_explore <input>
```

optional module
clustering
differential_expression

bundle field	value
enabled	True
include_protein	False
include_spatial	False
include_multimodal_obsm	False
schema_version	v2.1

When to use: Most common use case for new users on a single sample. Skips trajectory/annotation to keep wall-time low; rerun with another recipe (e.g. `nc2024_paper`) once you've inspected the bundle.

5.cite_seq_full

CITE-seq full analysis: RNA + ADT (CLR) + bundle with protein extension

Command:

```
python scripts/scfactory.py run --recipe cite_seq_full <input>
```

optional module
clustering
differential_expression
annotation
trajectory
protein_adt

bundle field	value
enabled	True
include_protein	True
include_spatial	False
include_multimodal_obsm	False
schema_version	v2.1

When to use: Pairs CITE-seq RNA + protein (ADT) with CLR-normalized protein outputs in the v2.1 bundle (protein extension on). Inputs may be a CITE-seq .h5ad with obsm['protein_counts'] or a cellranger sample-root with an Antibody Capture modality in the H5.

5.visium_neighborhoods

Visium spatial transcriptomics: ingest + neighborhood analytics + spatial bundle

Command:

```
python scripts/scfactory.py run --recipe visium_neighborhoods <input>
```

optional module
clustering
differential_expression
annotation
spatial_ingest
spatial_neighborhoods

bundle field	value
enabled	True
include_protein	False
include_spatial	True
include_multimodal_obsm	False
schema_version	v2.1

When to use: Visium / generic spatial workflow. `spatial_ingest` attaches (x,y) coords + library metadata; `spatial_neighborhoods` computes Moran's I, neighborhood enrichment, and co-occurrence. Requires squidpy in the active environment for full neighborhood analytics; without it, `spatial_neighborhoods` will degrade to a no-op with a warning. The v2.1 bundle's spatial extension is enabled.

6. scfactory CLI reference

scfactory

usage: scfactory [-h] {run,doctor,report} ...

User-friendly wrapper around the modular pipeline.

positional arguments:

{run,doctor,report}

run Auto-detect modality and dispatch to

workflow.modular.cli

doctor Read-only environment health check

report Render a single self-contained HTML report from a run

directory

options:

-h, --help show this help message and exit

scfactory run

usage: scfactory run [-h] [--project PROJECT] [--out OUT] [--bundle]

[--bundle-out BUNDLE_OUT]

[--optional-modules OPTIONAL_MODULES] [--recipe RECIPE]

[--list-recipes] [--dry-run]

[input]

Run the modular pipeline on an .h5ad or sample-root.

Module precedence (highest wins):

1. --optional-modules <list> (escape hatch; overrides all)

2. --recipe <name> (preset modules + env + bundle)

3. auto-detected modality (fills any remaining gap)

--bundle and recipe.bundle.enabled are OR-ed: passing --bundle always forces bundle on even if the recipe disables it.

positional arguments:

input .h5ad file or sample-root directory (omit only with

--list-recipes)

options:

-h, --help show this help message and exit

--project PROJECT Run name (default: scfactory_<modality>)

--out OUT Output directory (default: <repo>/results)

--bundle After run, export an R bundle via

scripts/export_singlecell_r_bundle.py (forces bundle on even if recipe disables it)

--bundle-out BUNDLE_OUT

Bundle output path (default: <out>/<project>/r_bundle)

--optional-modules OPTIONAL_MODULES

Override auto-detected optional module list (comma-

separated; passthrough escape hatch – wins over

--recipe)

--recipe RECIPE Apply a preset recipe from recipes/<name>.yaml

(modules + scale-mode + env + bundle config)

--list-recipes List available recipes (one per line) and exit 0

--dry-run Print what would run, do not execute

scfactory doctor

usage: scfactory doctor [-h] [--json]

options:

-h, --help show this help message and exit

--json Emit machine-readable JSON

scfactory report

usage: scfactory report [-h] [--out OUT] [--title TITLE] run_dir

positional arguments:

run_dir Path to a run directory containing run_manifest.json

options:

-h, --help show this help message and exit

--out OUT Output HTML path (default: <run_dir>/report.html)

--title TITLE Report title (default: project name from manifest)

7. NC2024 claim coverage

Source: multiomics_r_factory/docs/NSCLC_CLAIM_VALIDATION_MATRIX.md. **15** claims parsed: **4** direct, **7** partial, **4** unsupported.

claim_id	family	module_family	lane	status	evidence
NC2024-F1-01	broad cell-type atlas	core scRNA analysis	40k_fidelity	direct	40k clean run + annotation outputs
NC2024-F1-02	tumour/background composition shift	composition	40k_fidelity	direct	40k clean run composition outputs
NC2024-F1-03	immune composition shift	immune/TME	40k_fidelity	direct	40k clean immune outputs
NC2024-F1-04	Treg/NK/exhaustion pattern	immune/TME	40k_fidelity	direct	40k clean immune outputs
NC2024-F1-05	CAML-like hybrid state	scRNA-core / tumour-state	40k_fidelity	partial	40k marker/annotation outputs
NC2024-F2-01	abundance correlation	composition	40k_fidelity	partial	40k composition outputs
NC2024-F2-02	communication differences	cell-cell communication	none	unsupported	no current communication run
NC2024-F2-03	VEGF/EGFR interaction pattern	cell-cell communication	none	unsupported	no current communication run
NC2024-F2-04	LUSC-specific checkpoint wiring	cell-cell communication	none	unsupported	no current communication run
NC2024-F3-01	macrophage state diversity	immune/TME	900k_survivability	partial	900k proxy v3 immune + TME outputs
NC2024-F3-02	foetal-like macrophage programme	pathway/regulatory	none	unsupported	current runs lack dedicated pathway/regulatory analysis for this claim
NC2024-F4-01	spatial localization	spatial	bundle_v2.1	partial	AnnData -> bundle v2.1 `spatial` extension -> `R/spatial_module.R::plot_spatial...
NC2024-F4-02	spatial interaction context	spatial + communication	bundle_v2.1	partial	bundle v2.1 `spatial` extension + upstream `spatial_neighborhoods.py` (squidpy-...
NC2024-ABS-02	macrophage vs cytotoxic inverse relationship	immune/TME	900k_survivability	partial	900k immune subtype summary + composition + TME
NC2024-ABS-03	similar composition but subtype-specific divergence	subtype comparison	40k_fidelity	partial	40k clean + 40k benchmark + docs

8. Test coverage and quality gates

Total collected on the named contract surface: **97** tests.

file	tests	protects against
tests/test_r_bundle_contract.py	16	Python <-> R bundle v2/v2.1 round-trip subprocess contract; guards CLAIM_GUARD surfacing in R.
tests/test_python_r_parity.py	0	Cross-language sentinels (scanpy vs Seurat) — fires when defaults silently diverge.
tests/test_modular.py	54	Modular pipeline contracts: dependency resolution, ctx.set_module_dir, manifest/status outputs.
tests/test_scfactory.py	16	scfactory CLI wrapper: auto-detection planner, recipe precedence, doctor JSON shape.
tests/test_singlecell_r_bundle_export.py	5	Bundle export internals: schema versions, atomic publish, extension manifest entries.
tests/test_densify_audit.py	2	Grep-ban: every .toarray()/todense() in modules must carry a densify-allowed comment.
tests/test_module_catalog.py	4	Module catalog stability: mandatory chain, default optional list, layer coverage.

9. Known limitations and roadmap

- **multimodal_integration** is **EXPERIMENTAL**. Off by default, gated by `SC_MULTIMODAL_ENGINE`. Emits the bundle v2.1 *multimodal_obs* extension only when explicitly enabled.
- **Tissue images are STRING-pointer only** in the spatial extension. Image bytes are never serialized into the bundle; readers must resolve paths themselves.
- **Two bundle-v1 schema tests are pinned to v1** (not removed) to preserve back-compat coverage.
- **No cell-cell communication / pathway scoring / RNA-velocity / trajectory parity** yet against R; listed as roadmap.
- `--include-multimodal-obs` is opt-in only and is never auto-passed even on the multimodal modality.
- Pre-existing *repo-doc-sync* hook on `singlecell_factory` blocks commits until citation/doc references are staged.
- Roadmap: a dedicated "How to add a new module" extension guide is planned but is intentionally NOT included in this overview.

Appendix: file index

All top-level *.md and docs/*.md across both repos (24 entries).

path	summary
multiomics_r_factory/AGENTS.md	Start with `AI_AGENT_PROTOCOL.md` for onboarding, architecture, and task
multiomics_r_factory/AI_AGENT_PROTOCOL.md	Canonical onboarding index for AI agents entering
multiomics_r_factory/CLAUDE.md	Claude Code must start with `AI_AGENT_PROTOCOL.md` before editing this
multiomics_r_factory/CODEX_PROFILE.md	Codex must start with `AI_AGENT_PROTOCOL.md` for onboarding, architecture, and
multiomics_r_factory/MODULE_UPDATE_SKILL.md	Use this checklist every time a new module is added or an existing module is changed.
multiomics_r_factory/README.md	AI agents must start with AI_AGENT_PROTOCOL.md. That
multiomics_r_factory/SKILL_PLAYBOOK.md	Start with `AI_AGENT_PROTOCOL.md` before choosing a skill. This playbook routes
multiomics_r_factory/docs/ARTICLE_CLAIM_INVENTORY_NSCLC_NATCOMM_2024.md	Baseline article:
multiomics_r_factory/docs/MODULE_CAPABILITY_MAP.md	`singlecell_factory` is the compute and analysis engine. `multiomics_r_factory` is the downstream R/report wo...
multiomics_r_factory/docs/NC2024_NSCLC_BASELINE_REPRODUCTION_MEMO.md	- Journal: `Nature Communications`
multiomics_r_factory/docs/NSCLC_BASELINE_REPRO_TASK_LIST.md	This task list converts the baseline article into reproducible work items using the approved evidence schema.
multiomics_r_factory/docs/NSCLC_CLAIM_VALIDATION_MATRIX.md	This matrix is the claim authority table for the current baseline article.
multiomics_r_factory/docs/NSCLC_REPRODUCTION_COMPARISON_MEMO.md	This memo compares the three current NSCLC lanes and states what each lane proves.
multiomics_r_factory/docs/PYTHON_R_PROVENANCE_AND_FIDELITY.md	This document defines how to interpret Python-generated figures and R-generated figures in this workspace, an...
singlecell_factory/AGENTS.md	This file applies to `/home/zerlinshen/singlecell_factory` and all subdirectories.
singlecell_factory/AI_AGENT_PROTOCOL.md	Canonical onboarding index for AI agents entering
singlecell_factory/BEST_PRACTICES.md	- Codex ██████`AGENTS.md` ██████████
singlecell_factory/CLAUDE.md	Start with `AI_AGENT_PROTOCOL.md` for onboarding, read order, and task routing.
singlecell_factory/CODEX.md	Codex must start with `AI_AGENT_PROTOCOL.md` for onboarding, read order, and
singlecell_factory/PROTOCOL.md	AI agents should start with `AI_AGENT_PROTOCOL.md` before using this file. This
singlecell_factory/README.md	AI agents must start with AI_AGENT_PROTOCOL.md. That
singlecell_factory/docs/FACTORIES_OVERVIEW.md	_singlecell_factory + multiomics_r_factory technical reference_
singlecell_factory/docs/MODULE_TECH_DOC_TEMPLATE.md	- Module name:
singlecell_factory/docs/PUBLICATION_READY.md	This file provides copy-paste-ready Methods section text and citation patterns for manuscripts using this pip...